IMPERIAL COLLEGE LONDON

EE4-27
EE9-CS5-2
EE9-FPN2-08

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

SYSTEMS IDENTIFICATION

Wednesday, 13 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Parisini
Second Marker(s) : F. Teng

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SYSTEMS IDENTIFICATION

Recommendations for candidates:

The exam has to be carried out remotely following the official indications provided by the College. Moreover, the algebraic calculations have to be carried out using a standard calculator. The use of mathematical software such as Matlab and similar tools installed on personal computers is prohibited.

Information for candidates:

Consider an invertible 3 by 3 matrix A of the form

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

and denote by

$$A^{-1} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

its inverse.

Then, the first entry b_{11} of A^{-1} is given as follows:

$$b_{11} = \frac{a_{22}a_{33} - a_{23}a_{32}}{a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}}$$

1. Consider a stationary stochastic process $x(\cdot)$ satisfying the following difference equation:

$$x(t+1) = \frac{1}{2}x(t) + w(t)$$

where $w(\cdot) \sim WN(\bar{w}, 2)$ is a non-zero mean white stochastic process (that is, it is a white stochastic process to which an *unknown* constant $\bar{w}$ has been added). We assume that some samples of the stochastic process $x(\cdot)$ are collected, and the objective is to determine an estimate $\hat{S}$ of its expected value $\mathbb{E}[x(t)]$ from the measured samples.
Hint: since the stochastic process $x(\cdot)$ is stationary, then for every t_1 and t_2, it holds that $\mathbb{E}[x(t_1)] = \mathbb{E}[x(t_2)]$.

a) Let S_1, S_2, and S_3 denote three samples collected from the stochastic process $x(\cdot)$ at the three time instants $t = 1$, $t = 2$, and $t = 3$, respectively. Consider the family of linear estimators

$$\hat{S}(\alpha, \beta, \gamma) = \alpha S_1 + \beta S_2 + \gamma S_3$$

where $\alpha, \beta, \gamma \in \mathbb{R}$ are three parameters to be determined. Compute the set of all parameters α, β, γ leading to an unbiased estimator $\hat{S}(\alpha, \beta, \gamma)$.

[3 Marks]

b) Compute the expected value $\mathbb{E}[x(t)]$ and the variance $\text{var}[x(t)]$ of $x(t)$.

[4 Marks]

c) Compute the covariance matrix of the random vector $[x(1) \;\; x(2) \;\; x(3)]^\top$.

[5 Marks]

d) Find the parameters $(\alpha^\circ, \beta^\circ, \gamma^\circ)$ inside the set of α, β, γ determined in your answer to Question 1-a), such that $\text{var}[\hat{S}(\alpha^\circ, \beta^\circ, \gamma^\circ)]$ is the smallest possible.

[8 Marks]

2. Consider the stochastic system described by the block-scheme shown in Figure 2.1, where a_1 and a_2 are constant parameters. Moreover, $v(\cdot) \sim WN(1,1)$ is a non-zero mean white stochastic process (that is, it is a white stochastic process to which a constant has been added).

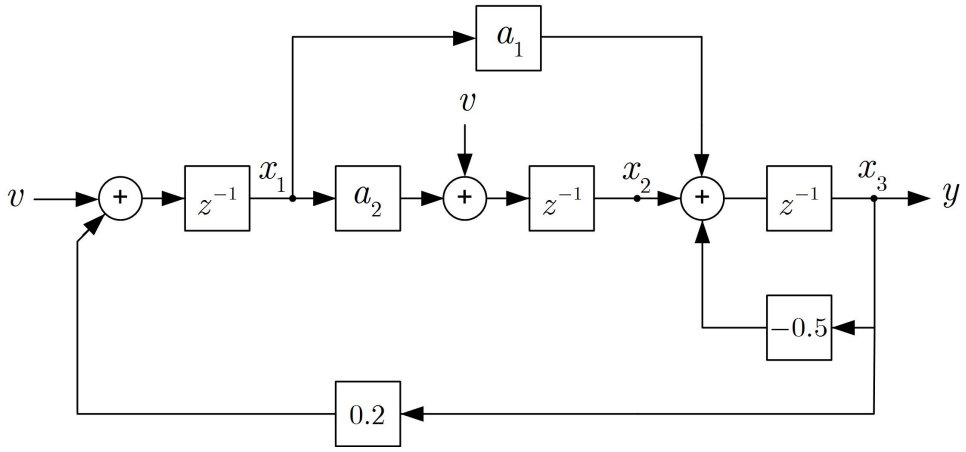


Figure 2.1 Block-scheme of the stochastic system of Question 2.

a) Find a suitable ARMA model (parameterised by a_1 and a_2) describing the stochastic process $y(\cdot)$.

[5 Marks]

b) Set $a_1 = 1$ and $a_2 = 0$. Show that the stochastic process $y(\cdot)$ is stationary.

[4 Marks]

c) Set $a_1 = 1$ and $a_2 = 0$, for which the stochastic processes $x_1(\cdot)$, $x_2(\cdot)$, and $x_3(\cdot)$ are stationary. Compute the expected values $m_1 = \mathbb{E}[x_1(t)]$, $m_2 = \mathbb{E}[x_2(t)]$, and $m_3 = \mathbb{E}[x_3(t)]$ of the stochastic processes $x_1(\cdot)$, $x_2(\cdot)$, and $x_3(\cdot)$, respectively.

[4 Marks]

d) Set $a_1 = 1$ and $a_2 = 0$. Compute the first three samples of the correlation function $\gamma_y(\tau), \tau = 0,1,2$, of the stochastic process $y(\cdot)$.

[7 Marks]

3. A person has a bank account in which he/she keeps his/her savings $S(t)$. At the end of each month t, the amount of savings is given by

$$S(t+1) = S(t) - C(t) + U(t) + I(t), \quad t \geq 1 \tag{3.1}$$

in which $U(t)$ is the monthly salary, $C(t)$ denotes the monthly expenses and it is given by $C(t) = \alpha S(t) + \beta(S(t) - S(t-1))^3 + \gamma U(t)$, and $I(\cdot) \sim WN(0, \sigma_I^2)$ is a white stochastic process describing possible model mismatches. Moreover, α, β, and γ are unknown constants. A data set of $M+1$ samples $\{(S(j), U(j)), j = 0, 1, 2, \ldots, M\}$, $M \geq 2$, has been collected in order to identify the model (3.1).

a) Write $S(t+1)$ as a function of $S(j)$ and $U(j)$, $j = t, t-1, \ldots$.

[3 Marks]

b) By considering the model obtained in your answer to Question 3-a), rewrite it in the form of a parametric model characterised as follows:

$$S(t+1) = \sum_{i=1}^{3} \theta_i p_i(t, t-1) + I(t), \quad t \geq 1$$

for a suitable choice of 3 known functions $p_i(t, t-1), i = 1, 2, 3$ and of a vector of unknown parameters $\theta = [\theta_1, \theta_2, \theta_3]^\top$.

[3 Marks]

c) Using the parametric model obtained in your answer to Question 3-b), devise a least-squares procedure to estimate the unknown vector of parameters θ, namely a procedure to find a $\hat{\theta} = [\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3]^\top$ minimising the following cost function:

$$J(\hat{\theta}) = \frac{1}{M-1} \sum_{j=1}^{M-1} \left(S(j+1) - \sum_{i=1}^{3} \hat{\theta}_i p_i(j, j-1) \right)^2 .$$

[6 Marks]

d) Which one of the following set of samples does guarantee that the procedure devised in your answer to Question 3-c) gives a unique solution for the vector of parameters $\hat{\theta}$? Justify your answer.

i) $M = 3$ and $(S(0), U(0)) = (1,2)$, $(S(1), U(1)) = (1,2)$, $(S(2), U(2)) = (2,2)$, $(S(3), U(3)) = (2.5, 2)$

ii) $M = 7$ and $(S(0), U(0)) = (1,2)$, $(S(1), U(1)) = (1,2)$, $(S(2), U(2)) = (2,2)$, $(S(3), U(3)) = (2.5, 2)$, $(S(4), U(4)) = (3,3)$, $(S(5), U(5)) = (4,3)$, $(S(6), U(6)) = (4.5, 3)$, $(S(7), U(7)) = (5,3)$.

[5 Marks]

e) For the sets of samples *i*) and *ii*) given in Question 3-d) compute, if possible, the variance $\text{var}[\hat{\theta}_1]$ for the parameter θ_1. Justify your answer.
Hint: in your calculations, please consider using the matrix inverse formula given in the "Information for candidates" section provided at the beginning of the exam paper.

[3 Marks]

4. With $v(\cdot)$ a stochastic process generated as

$$v(t) = e(t) - \frac{1}{2}e(t-1) - \frac{1}{2}e(t-2), \quad e(\cdot) \sim WN(0,1),$$

consider the following family of AR models:

$$\mathscr{M}(\theta): \quad v(t) = a_1 v(t-1) + a_2 v(t-2) + \xi(t),\ \theta = \begin{bmatrix} a_1 & a_2 \end{bmatrix}^\top,$$

where $\xi(\cdot)$ is a generic white process uncorrelated with $v(\cdot)$. Suppose an arbitrarily large number N of measurements $\{v(1), v(2), \ldots, v(N)\}$ has been collected, and denote by $\hat{\theta}(N) = \begin{bmatrix} \hat{a}_1(N) & \hat{a}_2(N) \end{bmatrix}^\top$ the least squares estimate of θ based on the N measurements $\{v(1), v(2), \ldots, v(N)\}$.

a) Determine the value $\bar{\theta}$ the estimate $\hat{\theta}$ approaches for large values of N (that is, $\bar{\theta} = \lim_{N\to\infty} \hat{\theta}$ a.s.).

[4 Marks]

b) For the value $\bar{\theta}$ obtained in your answer to Question 4-a), write the difference equation representing the optimal one-step ahead predictor $\hat{v}(t+1|t)$ of $v(t)$ from measured values of $v(\cdot)$.

[4 Marks]

c) For the value $\bar{\theta}$ obtained in your answer to Question 4-a), write the difference equation representing the optimal two-step ahead predictor $\hat{v}(t+2|t)$ of $v(t)$ from measured values of $v(\cdot)$.

[6 Marks]

d) For the optimal one and two-steps ahead predictors obtained in your answers to Questions 4-b) and 4-c), calculate the variance $\mathrm{var}[e_1(t)]$ and $\mathrm{var}[e_2(t)]$ of the prediction errors $e_1(t) = v(t+1) - \hat{v}(t+1|t)$ and $e_2(t) = v(t+2) - \hat{v}(t+2|t)$, respectively, and compare them. Discuss your findings.

[6 Marks]